

What is Unix?

The Unix operating system is a set of programs that act as a link between the computer and the user.

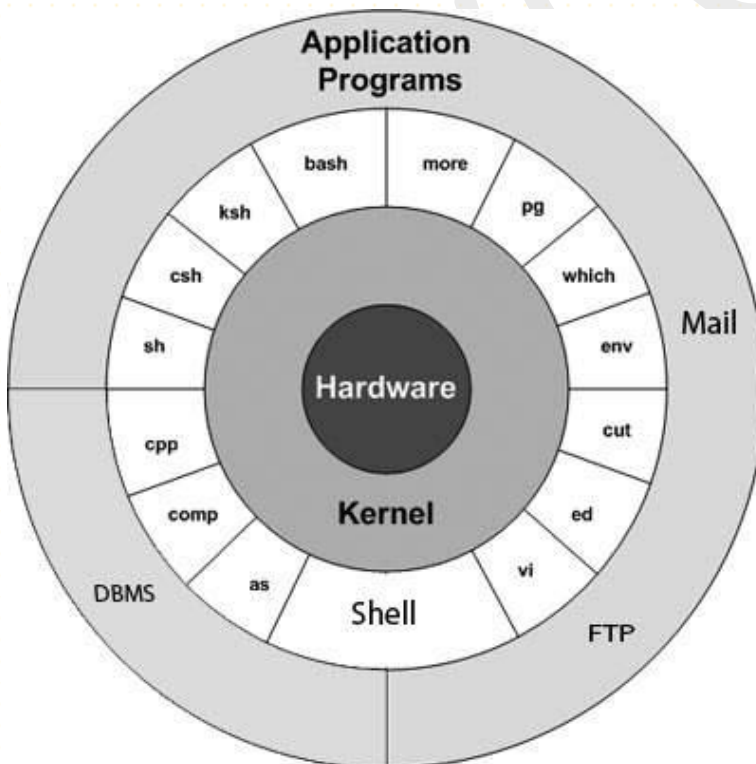
The computer programs that allocate the system resources and coordinate all the details of the computer's internals is called the **operating system** or the **kernel**.

Users communicate with the kernel through a program known as the **shell**. The shell is a command line interpreter; it translates commands entered by the user and converts them into a language that is understood by the kernel.

- Unix was originally developed in 1969 by a group of AT&T employees Ken Thompson, Dennis Ritchie, Douglas McIlroy, and Joe Ossanna at Bell Labs.
- There are various Unix variants available in the market. Solaris Unix, AIX, HP Unix and BSD are a few examples. Linux is also a flavor of Unix which is freely available.
- Several people can use a Unix computer at the same time; hence Unix is called a multiuser system.
- A user can also run multiple programs at the same time; hence Unix is a multitasking environment.

Unix Architecture

Here is a basic block diagram of a Unix system –



The main concept that unites all the versions of Unix is the following four basics –

- **Kernel** – The kernel is the heart of the operating system. It interacts with the hardware and most of the tasks like memory management, task scheduling and file management.
- **Shell** – The shell is the utility that processes your requests. When you type in a command at your terminal, the shell interprets the command and calls the program that you want. The shell uses standard syntax for all commands. C Shell, Bourne Shell and Korn Shell are the most famous shells which are available with most of the Unix variants.
- **Commands and Utilities** – There are various commands and utilities which you can make use of in your day to day activities. **cp**, **mv**, **cat** and **grep**, etc. are few examples of commands and utilities. There are over 250 standard commands plus numerous others provided through 3rd party software. All the commands come along with various options.
- **Files and Directories** – All the data of Unix is organized into files. All files are then organized into directories. These directories are further organized into a tree-like structure called the **filesystem**.

System Bootup

If you have a computer which has the Unix operating system installed in it, then you simply need to turn on the system to make it live.

As soon as you turn on the system, it starts booting up and finally it prompts you to log into the system, which is an activity to log into the system and use it for your day-to-day activities.

Login Unix

When you first connect to a Unix system, you usually see a prompt such as the following –

```
login:
```

To log in

- Have your userid (user identification) and password ready. Contact your system administrator if you don't have these yet.
- Type your userid at the login prompt, then press **ENTER**. Your userid is **case-sensitive**, so be sure you type it exactly as your system administrator has instructed.
- Type your password at the password prompt, then press **ENTER**. Your password is also case-sensitive.
- If you provide the correct userid and password, then you will be allowed to enter into the system. Read the information and messages that comes up on the screen, which is as follows.

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```
login : amrood
amrood's password:
Last login: Sun Jun 14 09:32:32 2009 from 62.61.164.73
$
```

You will be provided with a command prompt (sometime called the **\$** prompt) where you type all your commands. For example, to check calendar, you need to type the **cal** command as follows –

```
$ cal
   June 2009
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30
$
```

Change Password

All Unix systems require passwords to help ensure that your files and data remain your own and that the system itself is secure from hackers and crackers. Following are the steps to change your password –

Step 1 – To start, type password at the command prompt as shown below.

Step 2 – Enter your old password, the one you're currently using.

Step 3 – Type in your new password. Always keep your password complex enough so that nobody can guess it. But make sure, you remember it.

Step 4 – You must verify the password by typing it again.

```
$ passwd
Changing password for amrood
(current) Unix password:*****
New UNIX password:*****
Retype new UNIX password:*****
passwd: all authentication tokens updated successfully
$
```

Note – We have added asterisk (*) here just to show the location where you need to enter the current and new passwords otherwise at your system. It does not show you any character when you type.

Listing Directories and Files

All data in Unix is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.

You can use the **ls** command to list out all the files or directories available in a directory. Following is the example of using **ls** command with **-l** option.

```
$ ls -l
total 19621
drwxrwxr-x 2 amrood amrood  4096 Dec 25 09:59 uml
-rw-rw-r-- 1 amrood amrood  5341 Dec 25 08:38 uml.jpg
drwxr-xr-x 2 amrood amrood  4096 Feb 15  2006 univ
drwxr-xr-x 2 root  root    4096 Dec  9  2007 urlspedia
-rw-r--r-- 1 root  root   276480 Dec  9  2007 urlspedia.tar
drwxr-xr-x 8 root  root    4096 Nov 25  2007 usr
-rwxr-xr-x 1 root  root    3192 Nov 25  2007 webthumb.php
-rw-rw-r-- 1 amrood amrood  20480 Nov 25  2007 webthumb.tar
-rw-rw-r-- 1 amrood amrood   5654 Aug  9  2007 yourfile.mid
-rw-rw-r-- 1 amrood amrood 166255 Aug  9  2007 yourfile.swf
```

\$

Here entries starting with **d.....** represent directories. For example, uml, univ and urlspedia are directories and rest of the entries are files.

Who Are You?

While you're logged into the system, you might be willing to know : **Who am I?**

The easiest way to find out "who you are" is to enter the **whoami** command –

```
$ whoami
amrood
```

\$

Try it on your system. This command lists the account name associated with the current login. You can try **who am i** command as well to get information about yourself.

Who is Logged in?

Sometime you might be interested to know who is logged in to the computer at the same time.

There are three commands available to get you this information, based on how much you wish to know about the other users: **users**, **who**, and **w**.

```
$ users
```

```
amrood bablu qadir
```

```
$ who
```

```
amrood tty0 Oct 8 14:10 (limbo)
```

```
bablu tty2 Oct 4 09:08 (calliope)
```

```
qadir tty4 Oct 8 12:09 (dent)
```

```
$
```

Try the **w** command on your system to check the output. This lists down information associated with the users logged in the system.

Logging Out

When you finish your session, you need to log out of the system. This is to ensure that nobody else accesses your files.

To log out

- Just type the **logout** command at the command prompt, and the system will clean up everything and break the connection.

System Shutdown

The most consistent way to shut down a Unix system properly via the command line is to use one of the following commands –

S.No.	Command & Description
1	halt Brings the system down immediately
2	init 0 Powers off the system using predefined scripts to synchronize and clean up the system prior to shutting down
3	init 6 Reboots the system by shutting it down completely and then restarting it

4	poweroff Shuts down the system by powering off
5	reboot Reboots the system
6	shutdown Shuts down the system

You typically need to be the super user or root (the most privileged account on a Unix system) to shut down the system. However, on some standalone or personally-owned Unix boxes, an administrative user and sometimes regular users can do so.

Unix - File Management

In this chapter, we will discuss in detail about file management in Unix. All data in Unix is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.

When you work with Unix, one way or another, you spend most of your time working with files. This tutorial will help you understand how to create and remove files, copy and rename them, create links to them, etc.

In Unix, there are three basic types of files –

- **Ordinary Files** – An ordinary file is a file on the system that contains data, text, or program instructions. In this tutorial, you look at working with ordinary files.
- **Directories** – Directories store both special and ordinary files. For users familiar with Windows or Mac OS, Unix directories are equivalent to folders.
- **Special Files** – Some special files provide access to hardware such as hard drives, CD-ROM drives, modems, and Ethernet adapters. Other special files are similar to aliases or shortcuts and enable you to access a single file using different names.

Listing Files

To list the files and directories stored in the current directory, use the following command –

```
$ls
```

Here is the sample output of the above command –

```
$ls
bin    hosts lib  res.03
ch07   hw1  pub  test_results
ch07.bak hw2  res.01 users
docs   hw3  res.02 work
```

The command **ls** supports the **-l** option which would help you to get more information about the listed files –

```
$ls -l
total 1962188

drwxrwxr-x 2 amrood amrood  4096 Dec 25 09:59 uml
-rw-rw-r-- 1 amrood amrood  5341 Dec 25 08:38 uml.jpg
drwxr-xr-x 2 amrood amrood  4096 Feb 15  2006 univ
drwxr-xr-x 2 root  root    4096 Dec  9  2007 urlspedia
-rw-r--r-- 1 root  root   276480 Dec  9  2007 urlspedia.tar
drwxr-xr-x 8 root  root    4096 Nov 25  2007 usr
```

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```
drwxr-xr-x 2 200 300 4096 Nov 25 2007 webthumb-1.01
-rwxr-xr-x 1 root root 3192 Nov 25 2007 webthumb.php
-rw-rw-r-- 1 amrood amrood 20480 Nov 25 2007 webthumb.tar
-rw-rw-r-- 1 amrood amrood 5654 Aug 9 2007 yourfile.mid
-rw-rw-r-- 1 amrood amrood 166255 Aug 9 2007 yourfile.swf
drwxr-xr-x 11 amrood amrood 4096 May 29 2007 zlib-1.2.3
$
```

Here is the information about all the listed columns –

- **First Column** – Represents the file type and the permission given on the file. Below is the description of all type of files.
- **Second Column** – Represents the number of memory blocks taken by the file or directory.
- **Third Column** – Represents the owner of the file. This is the Unix user who created this file.
- **Fourth Column** – Represents the group of the owner. Every Unix user will have an associated group.
- **Fifth Column** – Represents the file size in bytes.
- **Sixth Column** – Represents the date and the time when this file was created or modified for the last time.
- **Seventh Column** – Represents the file or the directory name.

In the **ls -l** listing example, every file line begins with a **d**, **-**, or **l**. These characters indicate the type of the file that's listed.

S.No.	Prefix & Description
1	- Regular file, such as an ASCII text file, binary executable, or hard link.
2	b Block special file. Block input/output device file such as a physical hard drive.
3	c Character special file. Raw input/output device file such as a physical hard drive.
4	d Directory file that contains a listing of other files and directories.
5	l Symbolic link file. Links on any regular file.
6	p

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	Named pipe. A mechanism for interprocess communications.
7	s Socket used for interprocess communication.

Metacharacters

Metacharacters have a special meaning in Unix. For example, * and ? are metacharacters. We use * to match 0 or more characters, a question mark (?) matches with a single character.

For Example –

```
$ls ch*.doc
```

Displays all the files, the names of which start with **ch** and end with **.doc** –

```
ch01-1.doc ch010.doc ch02.doc ch03-2.doc
ch04-1.doc ch040.doc ch05.doc ch06-2.doc
ch01-2.doc ch02-1.doc c
```

Here, * works as meta character which matches with any character. If you want to display all the files ending with just **.doc**, then you can use the following command –

```
$ls *.doc
```

Hidden Files

An invisible file is one, the first character of which is the dot or the period character (.). Unix programs (including the shell) use most of these files to store configuration information.

Some common examples of the hidden files include the files –

- **.profile** – The Bourne shell (sh) initialization script
- **.kshrc** – The Korn shell (ksh) initialization script
- **.cshrc** – The C shell (csh) initialization script
- **.rhosts** – The remote shell configuration file

To list the invisible files, specify the **-a** option to **ls** –

```
$ ls -a
```

```
.      .profile  docs  lib   test_results
..     .rhosts   hosts  pub   users
.emacs bin      hw1    res.01 work
```

```
.exrc  ch07    hw2   res.02
.kshrc ch07.bak hw3   res.03
$
```

- **Single dot (.)** – This represents the current directory.
- **Double dot (..)** – This represents the parent directory.

Creating Files

You can use the **vi** editor to create ordinary files on any Unix system. You simply need to give the following command –

```
$ vi filename
```

The above command will open a file with the given filename. Now, press the key **i** to come into the edit mode. Once you are in the edit mode, you can start writing your content in the file as in the following program –

```
This is unix file....I created it for the first time.....
I'm going to save this content in this file.
```

Once you are done with the program, follow these steps –

- Press the key **esc** to come out of the edit mode.
- Press two keys **Shift + ZZ** together to come out of the file completely.

You will now have a file created with **filename** in the current directory.

```
$ vi filename
$
```

Editing Files

You can edit an existing file using the **vi** editor. We will discuss in short how to open an existing file –

```
$ vi filename
```

Once the file is opened, you can come in the edit mode by pressing the key **i** and then you can proceed by editing the file. If you want to move here and there inside a file, then first you need to come out of the edit mode by pressing the key **Esc**. After this, you can use the following keys to move inside a file –

- **l** key to move to the right side.
- **h** key to move to the left side.

- **k** key to move upside in the file.
- **j** key to move downside in the file.

So using the above keys, you can position your cursor wherever you want to edit. Once you are positioned, then you can use the **i** key to come in the edit mode. Once you are done with the editing in your file, press **Esc** and finally two keys **Shift + ZZ** together to come out of the file completely.

Display Content of a File

You can use the **cat** command to see the content of a file. Following is a simple example to see the content of the above created file –

```
$ cat filename
This is unix file....I created it for the first time.....
I'm going to save this content in this file.
$
```

You can display the line numbers by using the **-b** option along with the **cat** command as follows –

```
$ cat -b filename
1 This is unix file....I created it for the first time.....
2 I'm going to save this content in this file.
$
```

Counting Words in a File

You can use the **wc** command to get a count of the total number of lines, words, and characters contained in a file. Following is a simple example to see the information about the file created above –

```
$ wc filename
2 19 103 filename
$
```

Here is the detail of all the four columns –

- **First Column** – Represents the total number of lines in the file.
- **Second Column** – Represents the total number of words in the file.
- **Third Column** – Represents the total number of bytes in the file. This is the actual size of the file.
- **Fourth Column** – Represents the file name.

You can give multiple files and get information about those files at a time.

Following is simple syntax –

```
$ wc filename1 filename2 filename3
```

Copying Files

To make a copy of a file use the **cp** command. The basic syntax of the command is –

```
$ cp source_file destination_file
```

Following is the example to create a copy of the existing file **filename**.

```
$ cp filename copyfile  
$
```

You will now find one more file **copyfile** in your current directory. This file will exactly be the same as the original file **filename**.

Renaming Files

To change the name of a file, use the **mv** command. Following is the basic syntax –

```
$ mv old_file new_file
```

The following program will rename the existing file **filename** to **newfile**.

```
$ mv filename newfile  
$
```

The **mv** command will move the existing file completely into the new file. In this case, you will find only **newfile** in your current directory.

Deleting Files

To delete an existing file, use the **rm** command. Following is the basic syntax –

```
$ rm filename
```

Caution – A file may contain useful information. It is always recommended to be careful while using this **Delete** command. It is better to use the **-i** option along with **rm** command.

Following is the example which shows how to completely remove the existing file **filename**.

```
$ rm filename  
$
```

You can remove multiple files at a time with the command given below –

```
$ rm filename1 filename2 filename3
$
```

Standard Unix Streams

Under normal circumstances, every Unix program has three streams (files) opened for it when it starts up –

- **stdin** – This is referred to as the *standard input* and the associated file descriptor is 0. This is also represented as STDIN. The Unix program will read the default input from STDIN.
- **stdout** – This is referred to as the *standard output* and the associated file descriptor is 1. This is also represented as STDOUT. The Unix program will write the default output at STDOUT
- **stderr** – This is referred to as the *standard error* and the associated file descriptor is 2. This is also represented as STDERR. The Unix program will write all the error messages at STDERR.

Unix - Directory Management

In this chapter, we will discuss in detail about directory management in Unix. A directory is a file the solo job of which is to store the file names and the related information. All the files, whether ordinary, special, or directory, are contained in directories.

Unix uses a hierarchical structure for organizing files and directories. This structure is often referred to as a directory tree. The tree has a single root node, the slash character (/), and all other directories are contained below it.

Home Directory

The directory in which you find yourself when you first login is called your home directory.

You will be doing much of your work in your home directory and subdirectories that you'll be creating to organize your files.

You can go in your home directory anytime using the following command –

```
$cd ~  
$
```

Here ~ indicates the home directory. Suppose you have to go in any other user's home directory, use the following command –

```
$cd ~username  
$
```

To go in your last directory, you can use the following command –

```
$cd -  
$
```

Absolute/Relative Pathnames

Directories are arranged in a hierarchy with root (/) at the top. The position of any file within the hierarchy is described by its pathname.

Elements of a pathname are separated by a /. A pathname is absolute, if it is described in relation to root, thus absolute pathnames always begin with a /.

Following are some examples of absolute filenames.

```
/etc/passwd
```

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```
/users/sjones/chem/notes  
/dev/rdisk/Os3
```

A pathname can also be relative to your current working directory. Relative pathnames never begin with /. Relative to user amrood's home directory, some pathnames might look like this –

```
chem/notes  
personal/res
```

To determine where you are within the filesystem hierarchy at any time, enter the command **pwd** to print the current working directory –

```
$pwd  
/user0/home/amrood  
  
$
```

Listing Directories

To list the files in a directory, you can use the following syntax –

```
$ls dirname
```

Following is the example to list all the files contained in **/usr/local** directory –

```
$ls /usr/local  
  
X11    bin      gimp     jikes    sbin  
ace    doc      include  lib       share  
atalk  etc      info     man       ami
```

Creating Directories

We will now understand how to create directories. Directories are created by the following command –

```
$mkdir dirname
```

Here, directory is the absolute or relative pathname of the directory you want to create. For example, the command –

```
$mkdir mydir  
$
```

Creates the directory **mydir** in the current directory. Here is another example –

```
$mkdir /tmp/test-dir  
$
```

This command creates the directory **test-dir** in the **/tmp** directory. The **mkdir** command produces no output if it successfully creates the requested directory.

If you give more than one directory on the command line, **mkdir** creates each of the directories. For example, –

```
$mkdir docs pub
$
```

Creates the directories docs and pub under the current directory.

Creating Parent Directories

We will now understand how to create parent directories. Sometimes when you want to create a directory, its parent directory or directories might not exist. In this case, **mkdir** issues an error message as follows –

```
$mkdir /tmp/amrood/test
mkdir: Failed to make directory "/tmp/amrood/test";
No such file or directory
$
```

In such cases, you can specify the **-p** option to the **mkdir** command. It creates all the necessary directories for you. For example –

```
$mkdir -p /tmp/amrood/test
$
```

The above command creates all the required parent directories.

Removing Directories

Directories can be deleted using the **rmdir** command as follows –

```
$rmdir dirname
$
rm -rf
```

Note – To remove a directory, make sure it is empty which means there should not be any file or sub-directory inside this directory.

You can remove multiple directories at a time as follows –

```
$rmdir dirname1 dirname2 dirname3
$
```


The above command removes the directories `dirname1`, `dirname2`, and `dirname3`, if they are empty. The **rmdir** command produces no output if it is successful.

Changing Directories

You can use the **cd** command to do more than just change to a home directory. You can use it to change to any directory by specifying a valid absolute or relative path. The syntax is as given below –

```
$cd dirname
$
```

Here, **dirname** is the name of the directory that you want to change to. For example, the command –

```
$cd /usr/local/bin
$
```

Changes to the directory **/usr/local/bin**. From this directory, you can **cd** to the directory **/usr/home/amrood** using the following relative path –

```
$cd ../../home/amrood
$
```

Renaming Directories

The **mv (move)** command can also be used to rename a directory. The syntax is as follows –

```
$mv olddir newdir
$
```

You can rename a directory **mydir** to **yourdir** as follows –

```
$mv mydir yourdir
$
```

The directories **.** (dot) and **..** (dot dot)

The **filename .** (dot) represents the current working directory; and the **filename ..** (dot dot) represents the directory one level above the current working directory, often referred to as the parent directory.

If we enter the command to show a listing of the current working directories/files and use the **-a option** to list all the files and the **-l option** to provide the long listing, we will receive the following result.

```
$ls -la
drwxrwxr-x  4  teacher  class  2048 Jul 16 17:56 .
drwxr-xr-x 60   root      1536 Jul 13 14:18 ..
-----  1  teacher  class  4210 May 1 08:27 .profile
-rwxr-xr-x  1  teacher  class  1948 May 12 13:42 memo
```

Unix - File Permission / Access Modes

In this chapter, we will discuss in detail about file permission and access modes in Unix. File ownership is an important component of Unix that provides a secure method for storing files. Every file in Unix has the following attributes –

- **Owner permissions (u)**– The owner's permissions determine what actions the owner of the file can perform on the file.
- **Group permissions (g)**– The group's permissions determine what actions a user, who is a member of the group that a file belongs to, can perform on the file.
- **Other (world) permissions (o)**– The permissions for others indicate what action all other users can perform on the file.

The Permission Indicators

While using **ls -l** command, it displays various information related to file permission as follows –

```
$ls -l /home/amrood
-rwxr-xr-- 1 amrood  users 1024 Nov 2 00:10 myfile
drwxr-xr--- 1 amrood  users 1024 Nov 2 00:10 mydir
```

owner			group			others		
r	w	x	r	w	x	r	w	x
2	2	1	2	2	1	2	2	1
4+2+1 = 7			4+2+1			4+2+1		
4*0 + 2*0 + 1*0			0			0		

Here, the first column represents different access modes, i.e., the permission associated with a file or a directory.

The permissions are broken into groups of threes, and each position in the group denotes a specific permission, in this order: read (r), write (w), execute (x) –

- The first three characters (2-4) represent the permissions for the file's owner. For example, **-rwxr-xr--** represents that the owner has read (r), write (w) and execute (x) permission.
- The second group of three characters (5-7) consists of the permissions for the group to which the file belongs. For example, **-rwxr-xr--** represents that the group has read (r) and execute (x) permission, but no write permission.

- The last group of three characters (8-10) represents the permissions for everyone else. For example, **-rwxr-xr--** represents that there is **read (r)** only permission.

File Access Modes

The permissions of a file are the first line of defense in the security of a Unix system. The basic building blocks of Unix permissions are the **read**, **write**, and **execute** permissions, which have been described below –

Read: Grants the capability to read, i.e., view the contents of the file.

Write: Grants the capability to modify, or remove the content of the file.

Execute: User with execute permissions can run a file as a program.

Directory Access Modes

Directory access modes are listed and organized in the same manner as any other file. There are a few differences that need to be mentioned –

Read: Access to a directory means that the user can read the contents. The user can look at the filenames inside the directory.

Write: Access means that the user can add or delete files from the directory.

Execute: Executing a directory doesn't really make sense, so think of this as a traverse permission.

A user must have **execute** access to the **bin** directory in order to execute the **ls** or the **cd** command.

Changing Permissions

To change the file or the directory permissions, you use the **chmod** (change mode) command. There are two ways to use chmod — the symbolic mode and the absolute mode.

Using chmod in Symbolic Mode

The easiest way for a beginner to modify file or directory permissions is to use the symbolic mode. With symbolic permissions you can add, delete, or specify the permission set you want by using the operators in the following table.

S.No.	Chmod operator & Description
1	+

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	Adds the designated permission(s) to a file or directory.
2	- Removes the designated permission(s) from a file or directory.
3	= Sets the designated permission(s).

Here's an example using **testfile**. Running **ls -l** on the testfile shows that the file's permissions are as follows –

```
$ls -l testfile
-rwxrwxr-- 1 amrood  users 1024 Nov 2 00:10 testfile
```

Then each example **chmod** command from the preceding table is run on the testfile, followed by **ls -l**, so you can see the permission changes –

```
$chmod o+wx testfile
$ls -l testfile
-rwxrwxrwx 1 amrood  users 1024 Nov 2 00:10 testfile
$chmod u-x testfile
$ls -l testfile
-rw-rwxrwx 1 amrood  users 1024 Nov 2 00:10 testfile
$chmod g = rx testfile
$ls -l testfile
-rw-r-xrwx 1 amrood  users 1024 Nov 2 00:10 testfile
```

Here's how you can combine these commands on a single line –

```
$chmod o+wx,u-x,g = rx testfile
$ls -l testfile
-rw-r-xrwx 1 amrood  users 1024 Nov 2 00:10 testfile
```

Using chmod with Absolute Permissions

The second way to modify permissions with the **chmod** command is to use a number to specify each set of permissions for the file.

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Each permission is assigned a value, as the following table shows, and the total of each set of permissions provides a number for that set.

Number	Octal Permission Representation	Ref
0	No permission	---
1	Execute permission	--x
2	Write permission	-w-
3	Execute and write permission: 1 (execute) + 2 (write) = 3	-wx
4	Read permission	r--
5	Read and execute permission: 4 (read) + 1 (execute) = 5	r-x
6	Read and write permission: 4 (read) + 2 (write) = 6	rw-
7	All permissions: 4 (read) + 2 (write) + 1 (execute) = 7	rwX

Here's an example using the testfile. Running **ls -l** on the testfile shows that the file's permissions are as follows –

```
$ls -l testfile
-rwxrwxr-- 1 amrood  users 1024  Nov 2 00:10  testfile
```

Then each example **chmod** command from the preceding table is run on the testfile, followed by **ls -l**, so you can see the permission changes –

```
$ chmod 755 testfile
$ls -l testfile
-rwxr-xr-x 1 amrood  users 1024  Nov 2 00:10  testfile
$chmod 743 testfile
$ls -l testfile
-rwxr---wx 1 amrood  users 1024  Nov 2 00:10  testfile
$chmod 043 testfile
```

```
$ls -l testfile  
----r---wx 1 amrood  users 1024 Nov 2 00:10 testfile
```

Changing Owners and Groups

While creating an account on Unix, it assigns a **owner ID** and a **group ID** to each user. All the permissions mentioned above are also assigned based on the Owner and the Groups.

Two commands are available to change the owner and the group of files –

- **chown** – The **chown** command stands for "**change owner**" and is used to change the owner of a file.
- **chgrp** – The **chgrp** command stands for "**change group**" and is used to change the group of a file.

Changing Ownership

The **chown** command changes the ownership of a file. The basic syntax is as follows –

```
$ chown user filelist
```

The value of the user can be either the **name of a user** on the system or the **user id (uid)** of a user on the system.

The following example will help you understand the concept –

```
$ chown amrood testfile  
$
```

Changes the owner of the given file to the user **amrood**.

NOTE – The super user, root, has the unrestricted capability to change the ownership of any file but normal users can change the ownership of only those files that they own.

Changing Group Ownership

The **chgrp** command changes the group ownership of a file. The basic syntax is as follows –

```
$ chgrp group filelist
```

The value of group can be the **name of a group** on the system or the **group ID (GID)** of a group on the system.

Following example helps you understand the concept –

```
$ chgrp special testfile  
$
```

Changes the group of the given file to **special** group.

SUID and SGID File Permission

Often when a command is executed, it will have to be executed with special privileges in order to accomplish its task.

As an example, when you change your password with the **passwd** command, your new password is stored in the file **/etc/shadow**.

As a regular user, you do not have **read** or **write** access to this file for security reasons, but when you change your password, you need to have the write permission to this file. This means that the **passwd** program has to give you additional permissions so that you can write to the file **/etc/shadow**.

Additional permissions are given to programs via a mechanism known as the **Set User ID (SUID)** and **Set Group ID (SGID)** bits.

When you execute a program that has the SUID bit enabled, you inherit the permissions of that program's owner. Programs that do not have the SUID bit set are run with the permissions of the user who started the program.

This is the case with SGID as well. Normally, programs execute with your group permissions, but instead your group will be changed just for this program to the group owner of the program.

The SUID and SGID bits will appear as the letter **"s"** if the permission is available. The SUID **"s"** bit will be located in the permission bits where the owners' **execute** permission normally resides.

For example, the command –

```
$ ls -l /usr/bin/passwd  
-r-sr-xr-x 1 root bin 19031 Feb 7 13:47 /usr/bin/passwd*  
$
```

Shows that the SUID bit is set and that the command is owned by the root. A capital letter **S** in the execute position instead of a lowercase **s** indicates that the execute bit is not set.

If the sticky bit is enabled on the directory, files can only be removed if you are one of the following users –

- The owner of the sticky directory
- The owner of the file being removed

- The super user, root

To set the SUID and SGID bits for any directory try the following command –

```
$ chmod ug+s dirname  
$ ls -l  
drwsr-sr-x 2 root root 4096 Jun 19 06:45 dirname  
$
```


Unix – Environment

In this chapter, we will discuss in detail about the Unix environment. An important Unix concept is the **environment**, which is defined by environment variables. Some are set by the system, others by you, yet others by the shell, or any program that loads another program.

A variable is a character string to which we assign a value. The value assigned could be a number, text, filename, device, or any other type of data.

For example, first we set a variable TEST and then we access its value using the **echo** command –

```
$TEST="Unix Programming"
$echo $TEST
```

It produces the following result.

```
Unix Programming
```

Note that the environment variables are set without using the \$ sign but while accessing them we use the \$ sign as prefix. These variables retain their values until we come out of the shell.

When you log in to the system, the shell undergoes a phase called **initialization** to set up the environment. This is usually a two-step process that involves the shell reading the following files –

- /etc/profile
- profile

The process is as follows –

- The shell checks to see whether the file **/etc/profile** exists.
- If it exists, the shell reads it. Otherwise, this file is skipped. No error message is displayed.
- The shell checks to see whether the file **.profile** exists in your home directory. Your home directory is the directory that you start out in after you log in.
- If it exists, the shell reads it; otherwise, the shell skips it. No error message is displayed.

As soon as both of these files have been read, the shell displays a prompt –

```
$
```

This is the prompt where you can enter commands in order to have them executed.

Note – The shell initialization process detailed here applies to all **Bourne** type shells, but some additional files are used by **bash** and **ksh**.

The .profile File

The file **/etc/profile** is maintained by the system administrator of your Unix machine and contains shell initialization information required by all users on a system.

The file **.profile** is under your control. You can add as much shell customization information as you want to this file. The minimum set of information that you need to configure includes –

- The type of terminal you are using.
- A list of directories in which to locate the commands.
- A list of variables affecting the look and feel of your terminal.

You can check your **.profile** available in your home directory. Open it using the vi editor and check all the variables set for your environment.

Setting the Terminal Type

Usually, the type of terminal you are using is automatically configured by either the **login** or **getty** programs. Sometimes, the auto configuration process guesses your terminal incorrectly.

If your terminal is set incorrectly, the output of the commands might look strange, or you might not be able to interact with the shell properly.

To make sure that this is not the case, most users set their terminal to the lowest common denominator in the following way –

```
$TERM=vt100
$
```

Setting the PATH

When you type any command on the command prompt, the shell has to locate the command before it can be executed.

The PATH variable specifies the locations in which the shell should look for commands. Usually the Path variable is set as follows –

```
$PATH=/bin:/usr/bin
```

```
$
```

Here, each of the individual entries separated by the colon character (:) are directories. If you request the shell to execute a command and it cannot find it in any of the directories given in the PATH variable, a message similar to the following appears –

```
$hello
hello: not found
$
```

There are variables like PS1 and PS2 which are discussed in the next section.

PS1 and PS2 Variables

The characters that the shell displays as your command prompt are stored in the variable PS1. You can change this variable to be anything you want. As soon as you change it, it'll be used by the shell from that point on.

For example, if you issued the command –

```
$PS1='=>'
=>
=>
=>
```

Your prompt will become =>. To set the value of **PS1** so that it shows the working directory, issue the command –

```
=>PS1="\u@\h \w]\$"
[root@ip-72-167-112-17 /var/www/tutorialspoint/unix]$
[root@ip-72-167-112-17 /var/www/tutorialspoint/unix]$
```

The result of this command is that the prompt displays the user's username, the machine's name (hostname), and the working directory.

There are quite a few **escape sequences** that can be used as value arguments for PS1; try to limit yourself to the most critical so that the prompt does not overwhelm you with information.

S.No.	Escape Sequence & Description
1	\t

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	Current time, expressed as HH:MM:SS
2	<code>\d</code> Current date, expressed as Weekday Month Date
3	<code>\n</code> Newline
4	<code>\s</code> Current shell environment
5	<code>\w</code> Working directory
6	<code>\w</code> Full path of the working directory
7	<code>\u</code> Current user's username
8	<code>\h</code> Hostname of the current machine
9	<code>\#</code> Command number of the current command. Increases when a new command is entered

10

\\$

If the effective UID is 0 (that is, if you are logged in as root), end the prompt with the # character; otherwise, use the \$ sign

You can make the change yourself every time you log in, or you can have the change made automatically in PS1 by adding it to your **.profile** file.

When you issue a command that is incomplete, the shell will display a secondary prompt and wait for you to complete the command and hit **Enter** again.

The default secondary prompt is > (the greater than sign), but can be changed by re-defining the **PS2** shell variable –

Following is the example which uses the default secondary prompt –

```
$ echo "this is a  
> test"  
this is a  
test  
$
```

The example given below re-defines PS2 with a customized prompt –

```
$ PS2="secondary prompt->"  
$ echo "this is a  
secondary prompt->test"  
this is a  
test  
$
```

Environment Variables

Following is the partial list of important environment variables. These variables are set and accessed as mentioned below –

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S.No.	Variable & Description
1	DISPLAY Contains the identifier for the display that X11 programs should use by default.
2	HOME Indicates the home directory of the current user: the default argument for the cd built-in command.
3	IFS Indicates the Internal Field Separator that is used by the parser for word splitting after expansion.
4	LANG LANG expands to the default system locale; LC_ALL can be used to override this. For example, if its value is pt_BR , then the language is set to (Brazilian) Portuguese and the locale to Brazil.
5	LD_LIBRARY_PATH A Unix system with a dynamic linker, contains a colon-separated list of directories that the dynamic linker should search for shared objects when building a process image after exec, before searching in any other directories.
6	PATH Indicates the search path for commands. It is a colon-separated list of directories in which the shell looks for commands.
7	PWD Indicates the current working directory as set by the cd command.

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8	RANDOM Generates a random integer between 0 and 32,767 each time it is referenced.
9	SHLVL Increments by one each time an instance of bash is started. This variable is useful for determining whether the built-in exit command ends the current session.
10	TERM Refers to the display type.
11	TZ Refers to Time zone. It can take values like GMT, AST, etc.
12	UID Expands to the numeric user ID of the current user, initialized at the shell startup.

Following is the sample example showing few environment variables –

```
$ echo $HOME
/root
]$ echo $DISPLAY

$ echo $TERM
xterm
$ echo $PATH
/usr/local/bin:/bin:/usr/bin:/home/amrood/bin:/usr/local/bin
$
```

Unix - Pipes and Filters

In this chapter, we will discuss in detail about pipes and filters in Unix. You can connect two commands together so that the output from one program becomes the input of the next program. Two or more commands connected in this way form a pipe.

To make a pipe, put a vertical bar (|) on the command line between two commands.

When a program takes its input from another program, it performs some operation on that input, and writes the result to the standard output. It is referred to as a **filter**.

The grep Command

The grep command searches a file or files for lines that have a certain pattern. The syntax is –

```
$grep pattern file(s)
```

The name "**grep**" comes from the ed (a Unix line editor) command **g/re/p** which means “globally search for a regular expression and print all lines containing it”.

A regular expression is either some plain text (a word, for example) and/or special characters used for pattern matching.

The simplest use of grep is to look for a pattern consisting of a single word. It can be used in a pipe so that only those lines of the input files containing a given string are sent to the standard output. If you don't give grep a filename to read, it reads its standard input; that's the way all filter programs work –

```
$ls -l | grep "Aug"
-rw-rw-rw- 1 john doc 11008 Aug 6 14:10 ch02
-rw-rw-rw- 1 john doc 8515 Aug 6 15:30 ch07
-rw-rw-r-- 1 john doc 2488 Aug 15 10:51 intro
-rw-rw-r-- 1 carol doc 1605 Aug 23 07:35 macros
$
```

There are various options which you can use along with the **grep** command –

S.No.	Option & Description
1	-v Prints all lines that do not match pattern.

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2	-n Prints the matched line and its line number.
3	-l Prints only the names of files with matching lines (letter "l")
4	-c Prints only the count of matching lines.
5	-i Matches either upper or lowercase.

Let us now use a regular expression that tells `grep` to find lines with "**carol**", followed by zero or other characters abbreviated in a regular expression as `".*")`, then followed by "**Aug**".–

Here, we are using the `-i` option to have case insensitive search –

```
$ls -l | grep -i "carol.*aug"
-rw-rw-r-- 1 carol doc    1605 Aug 23 07:35 macros
$
```

The sort Command

The **sort** command arranges lines of text alphabetically or numerically. The following example sorts the lines in the food file –

```
$sort food
Afghani Cuisine
Bangkok Wok
Big Apple Deli
Isle of Java

Mandalay
Sushi and Sashimi
Sweet Tooth
Tio Pepe's Peppers
$
```

The **sort** command arranges lines of text alphabetically by default. There are many options that control the sorting –

S.No.	Description
1	-n Sorts numerically (example: 10 will sort after 2), ignores blanks and tabs.
2	-r Reverses the order of sort.
3	-f Sorts upper and lowercase together.
4	+x Ignores first x fields when sorting.

More than two commands may be linked up into a pipe. Taking a previous pipe example using **grep**, we can further sort the files modified in August by the order of size.

The following pipe consists of the commands **ls**, **grep**, and **sort** –

```
$ls -l | grep "Aug" | sort +4n
-rw-rw-r-- 1 carol doc    1605 Aug 23 07:35 macros
-rw-rw-r-- 1 john  doc    2488 Aug 15 10:51 intro
-rw-rw-rw- 1 john  doc    8515 Aug  6 15:30 ch07
-rw-rw-rw- 1 john  doc   11008 Aug  6 14:10 ch02
$
```

This pipe sorts all files in your directory modified in August by the order of size, and prints them on the terminal screen. The sort option **+4n** skips four fields (fields are separated by blanks) then sorts the lines in numeric order.

The **pg** and more Commands

A long output can normally be zipped by you on the screen, but if you run text through more or use the **pg** command as a filter; the display stops once the screen is full of text.

Let's assume that you have a long directory listing. To make it easier to read the sorted listing, pipe the output through **more** as follows –

```
$ls -l | grep "Aug" | sort +4n | more
-rw-rw-r-- 1 carol doc    1605 Aug 23 07:35 macros
-rw-rw-r-- 1 john  doc    2488 Aug 15 10:51 intro
-rw-rw-rw- 1 john  doc    8515 Aug  6 15:30 ch07
-rw-rw-r-- 1 john  doc   14827 Aug  9 12:40 ch03
.
.
.
-rw-rw-rw- 1 john  doc   16867 Aug  6 15:56 ch05
--More--(74%)
```

The screen will fill up once the screen is full of text consisting of lines sorted by the order of the file size. At the bottom of the screen is the **more** prompt, where you can type a command to move through the sorted text.

Once you're done with this screen, you can use any of the commands listed in the discussion of the more program.

Unix - Processes Management

In this chapter, we will discuss in detail about process management in Unix. When you execute a program on your Unix system, the system creates a special environment for that program. This environment contains everything needed for the system to run the program as if no other program were running on the system.

Whenever you issue a command in Unix, it creates, or starts, a new process. When you tried out the **ls** command to list the directory contents, you started a process. A process, in simple terms, is an instance of a running program.

The operating system tracks processes through a five-digit ID number known as the **pid** or the **process ID**. Each process in the system has a unique **pid**.

Pids eventually repeat because all the possible numbers are used up and the next pid rolls or starts over. At any point of time, no two processes with the same pid exist in the system because it is the pid that Unix uses to track each process.

Starting a Process

When you start a process (run a command), there are two ways you can run it –

- Foreground Processes
- Background Processes

Foreground Processes

By default, every process that you start runs in the foreground. It gets its input from the keyboard and sends its output to the screen.

You can see this happen with the **ls** command. If you wish to list all the files in your current directory, you can use the following command –

```
$ls ch*.doc
```

This would display all the files, the names of which start with **ch** and end with **.doc** –

```
ch01-1.doc ch010.doc ch02.doc ch03-2.doc  
ch04-1.doc ch040.doc ch05.doc ch06-2.doc  
ch01-2.doc ch02-1.doc
```

The process runs in the foreground, the output is directed to my screen, and if the **ls** command wants any input (which it does not), it waits for it from the keyboard.

While a program is running in the foreground and is time-consuming, no other commands can be run (start any other processes) because the prompt would not be available until the program finishes processing and comes out.

Background Processes

A background process runs without being connected to your keyboard. If the background process requires any keyboard input, it waits.

The advantage of running a process in the background is that you can run other commands; you do not have to wait until it completes to start another!

The simplest way to start a background process is to add an ampersand (&) at the end of the command.

```
$ls ch*.doc &
```

This displays all those files the names of which start with **ch** and end with **.doc** –

```
ch01-1.doc ch010.doc ch02.doc ch03-2.doc
ch04-1.doc ch040.doc ch05.doc ch06-2.doc
ch01-2.doc ch02-1.doc
```

Here, if the **ls** command wants any input (which it does not), it goes into a stop state until we move it into the foreground and give it the data from the keyboard.

That first line contains information about the background process - the job number and the process ID. You need to know the job number to manipulate it between the background and the foreground.

Press the Enter key and you will see the following –

```
[1] + Done      ls ch*.doc &
$
```

The first line tells you that the **ls** command background process finishes successfully. The second is a prompt for another command.

Listing Running Processes

It is easy to see your own processes by running the **ps** (process status) command as follows –

```
$ps
PID  TTY  TIME  CMD
18358 ttyp3 00:00:00 sh
18361 ttyp3 00:01:31 abiword
18789 ttyp3 00:00:00 ps
```

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One of the most commonly used flags for `ps` is the `-f` (`f` for full) option, which provides more information as shown in the following example –

```
$ps -f
UID   PID PPID C STIME  TTY  TIME CMD
amrood 6738 3662 0 10:23:03 pts/6 0:00 first_one
amrood 6739 3662 0 10:22:54 pts/6 0:00 second_one
amrood 3662 3657 0 08:10:53 pts/6 0:00 -ksh
amrood 6892 3662 4 10:51:50 pts/6 0:00 ps -f
```

Here is the description of all the fields displayed by `ps -f` command –

S.No.	Column & Description
1	UID User ID that this process belongs to (the person running it)
2	PID Process ID
3	PPID Parent process ID (the ID of the process that started it)
4	C CPU utilization of process
5	STIME Process start time
6	TTY Terminal type associated with the process

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7	TIME CPU time taken by the process
8	CMD The command that started this process

There are other options which can be used along with **ps** command –

S.No.	Option & Description
1	-a Shows information about all users
2	-x Shows information about processes without terminals
3	-u Shows additional information like -f option
4	-e Displays extended information

Stopping Processes

Ending a process can be done in several different ways. Often, from a console-based command, sending a CTRL + C keystroke (the default interrupt character) will exit the command. This works when the process is running in the foreground mode.

If a process is running in the background, you should get its Job ID using the **ps** command. After that, you can use the **kill** command to kill the process as follows –

```
$ps -f
UID    PID PPID C STIME  TTY  TIME CMD
amrood 6738 3662 0 10:23:03 pts/6 0:00 first_one
amrood 6739 3662 0 10:22:54 pts/6 0:00 second_one
amrood 3662 3657 0 08:10:53 pts/6 0:00 -ksh
amrood 6892 3662 4 10:51:50 pts/6 0:00 ps -f
$kill 6738
Terminated
```

Here, the **kill** command terminates the **first_one** process. If a process ignores a regular kill command, you can use **kill -9** followed by the process ID as follows –

```
$kill -9 6738
Terminated
```

Parent and Child Processes

Each unix process has two ID numbers assigned to it: The Process ID (pid) and the Parent process ID (ppid). Each user process in the system has a parent process.

Most of the commands that you run have the shell as their parent. Check the **ps -f** example where this command listed both the process ID and the parent process ID.

Zombie and Orphan Processes

Normally, when a child process is killed, the parent process is updated via a **SIGCHLD** signal. Then the parent can do some other task or restart a new child as needed. However, sometimes the parent process is killed before its child is killed. In this case, the "parent of all processes," the **init** process, becomes the new PPID (parent process ID). In some cases, these processes are called orphan processes.

When a process is killed, a **ps** listing may still show the process with a **Z** state. This is a zombie or defunct process. The process is dead and not being used. These processes are different from the orphan processes. They have completed execution but still find an entry in the process table.

Daemon Processes

Daemons are system-related background processes that often run with the permissions of root and services requests from other processes.

A daemon has no controlling terminal. It cannot open **/dev/tty**. If you do a "**ps -ef**" and look at the **tty** field, all daemons will have a **?** for the **tty**.

To be precise, a daemon is a process that runs in the background, usually waiting for something to happen that it is capable of working with. For example, a printer daemon waiting for print commands.

If you have a program that calls for lengthy processing, then it's worth to make it a daemon and run it in the background.

The top Command

The **top** command is a very useful tool for quickly showing processes sorted by various criteria.

It is an interactive diagnostic tool that updates frequently and shows information about physical and virtual memory, CPU usage, load averages, and your busy processes.

Here is the simple syntax to run top command and to see the statistics of CPU utilization by different processes –

```
$top
```

Job ID Versus Process ID

Background and suspended processes are usually manipulated via **job number (job ID)**. This number is different from the process ID and is used because it is shorter.

In addition, a job can consist of multiple processes running in a series or at the same time, in parallel. Using the job ID is easier than tracking individual processes.

VI EDITOR

In this chapter, we will understand how the vi Editor works in Unix. There are many ways to edit files in Unix. Editing files using the screen-oriented text editor **vi** is one of the best ways. This editor enables you to edit lines in context with other lines in the file.

An improved version of the vi editor which is called the **VIM** has also been made available now. Here, VIM stands for **Vi IM**proved.

vi is generally considered the de facto standard in Unix editors because –

- It's usually available on all the flavors of Unix system.
- Its implementations are very similar across the board.
- It requires very few resources.
- It is more user-friendly than other editors such as the **ed** or the **ex**.

You can use the **vi** editor to edit an existing file or to create a new file from scratch. You can also use this editor to just read a text file.

Starting the vi Editor

The following table lists out the basic commands to use the vi editor –

S.No.	Command & Description
1	vi filename Creates a new file if it already does not exist, otherwise opens an existing file.
2	vi -R filename Opens an existing file in the read-only mode.
3	view filename Opens an existing file in the read-only mode.

Following is an example to create a new file **testfile** if it already does not exist in the current working directory –

```
$vi testfile
```

The above command will generate the following output –

You will notice a **tilde** (~) on each line following the cursor. A tilde represents an unused line. If a line does not begin with a tilde and appears to be blank, there is a space, tab, newline, or some other non-viewable character present.

You now have one open file to start working on. Before proceeding further, let us understand a few important concepts.

Operation Modes

While working with the vi editor, we usually come across the following two modes –

- **Command mode** – This mode enables you to perform administrative tasks such as saving the files, executing the commands, moving the cursor, cutting (yanking) and pasting the lines or words, as well as finding and replacing. In this mode, whatever you type is interpreted as a command.
- **Insert mode** – This mode enables you to insert text into the file. Everything that's typed in this mode is interpreted as input and placed in the file.

vi always starts in the **command mode**. To enter text, you must be in the insert mode for which simply type **i**. To come out of the insert mode, press the **Esc** key, which will take you back to the command mode.

Hint – If you are not sure which mode you are in, press the Esc key twice; this will take you to the command mode. You open a file using the vi editor. Start by typing some characters and then come to the command mode to understand the difference.

Getting Out of vi

The command to quit out of vi is **:q**. Once in the command mode, type colon, and 'q', followed by return. If your file has been modified in any way, the editor will warn you of this, and not let you quit. To ignore this message, the command to quit out of vi without saving is **:q!**. This lets you exit vi without saving any of the changes.

The command to save the contents of the editor is **:w**. You can combine the above command with the quit command, or use **:wq** and return.

The easiest way to **save your changes and exit vi** is with the ZZ command. When you are in the command mode, type **ZZ**. The **ZZ** command works the same way as the **:wq** command.

If you want to specify/state any particular name for the file, you can do so by specifying it after the **:w**. For example, if you wanted to save the file you were working on as another filename called **filename2**, you would type **:w filename2** and return.

Moving within a File

To move around within a file without affecting your text, you must be in the command mode (press Esc twice). The following table lists out a few commands you can use to move around one character at a time

–

S.No.	Command & Description
1	k Moves the cursor up one line
2	j Moves the cursor down one line
3	h Moves the cursor to the left one character position

4	 Moves the cursor to the right one character position
---	--

The following points need to be considered to move within a file –

- vi is case-sensitive. You need to pay attention to capitalization when using the commands.
- Most commands in vi can be prefaced by the number of times you want the action to occur. For example, **2j** moves the cursor two lines down the cursor location.

There are many other ways to move within a file in vi. Remember that you must be in the command mode (**press Esc twice**). The following table lists out a few commands to move around the file –

Given below is the list of commands to move around the file.

Control Commands

The following commands can be used with the Control Key to performs functions as given in the table below –

Given below is the list of control commands.

Editing Files

To edit the file, you need to be in the insert mode. There are many ways to enter the insert mode from the command mode –

S.No.	Command & Description
1	i Inserts text before the current cursor location
2	I Inserts text at the beginning of the current line
3	a

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		Inserts text after the current cursor location
4	A	Inserts text at the end of the current line
5	o	Creates a new line for text entry below the cursor location
6	O	Creates a new line for text entry above the cursor location

Deleting Characters

Here is a list of important commands, which can be used to delete characters and lines in an open file –

S.No.	Command & Description	
1	x	Deletes the character under the cursor location
2	X	Deletes the character before the cursor location
3	dw	Deletes from the current cursor location to the next word
4	d^	Deletes from the current cursor position to the beginning of the line

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5	d\$ Deletes from the current cursor position to the end of the line
6	D Deletes from the cursor position to the end of the current line
7	dd Deletes the line the cursor is on

As mentioned above, most commands in vi can be prefaced by the number of times you want the action to occur. For example, **2x** deletes two characters under the cursor location and **2dd** deletes two lines the cursor is on.

It is recommended that the commands are practiced before we proceed further.

Change Commands

You also have the capability to change characters, words, or lines in vi without deleting them. Here are the relevant commands –

S.No.	Command & Description
1	cc Removes the contents of the line, leaving you in insert mode.
2	cw Changes the word the cursor is on from the cursor to the lowercase w end of the word.
3	r Replaces the character under the cursor. vi returns to the command mode after the replacement is entered.

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4	R Overwrites multiple characters beginning with the character currently under the cursor. You must use Esc to stop the overwriting.
5	s Replaces the current character with the character you type. Afterward, you are left in the insert mode.
6	S Deletes the line the cursor is on and replaces it with the new text. After the new text is entered, vi remains in the insert mode.

Copy and Paste Commands

You can copy lines or words from one place and then you can paste them at another place using the following commands –

S.No.	Command & Description
1	yy Copies the current line.
2	yw Copies the current word from the character the lowercase w cursor is on, until the end of the word.
3	p Puts the copied text after the cursor.

4

P

Puts the yanked text before the cursor.

Advanced Commands

There are some advanced commands that simplify day-to-day editing and allow for more efficient use of vi –

Given below is the list advanced commands.

Word and Character Searching

The vi editor has two kinds of searches: **string** and **character**. For a string search, the **/** and **?** commands are used. When you start these commands, the command just typed will be shown on the last line of the screen, where you type the particular string to look for.

These two commands differ only in the direction where the search takes place –

- The **/** command searches forwards (downwards) in the file.
- The **?** command searches backwards (upwards) in the file.

The **n** and **N** commands repeat the previous search command in the same or the opposite direction, respectively. Some characters have special meanings. These characters must be preceded by a backslash (****) to be included as part of the search expression.

S.No.	Character &Description
1	^ Searches at the beginning of the line (Use at the beginning of a search expression).
2	. Matches a single character.

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3	* Matches zero or more of the previous character.
4	\$ End of the line (Use at the end of the search expression).
5	[Starts a set of matching or non-matching expressions.
6	< This is put in an expression escaped with the backslash to find the ending or the beginning of a word.
7	> This helps see the '<' character description above.

The character search searches within one line to find a character entered after the command. The **f** and **F** commands search for a character on the current line only. **f** searches forwards and **F** searches backwards and the cursor moves to the position of the found character.

The **t** and **T** commands search for a character on the current line only, but for **t**, the cursor moves to the position before the character, and **T** searches the line backwards to the position after the character.

Set Commands

You can change the look and feel of your vi screen using the following **:set** commands. Once you are in the command mode, type **:set** followed by any of the following commands.

S.No.	Command & Description
1	:set ic Ignores the case when searching

2	:set ai Sets autoindent
3	:set noai Unsets autoindent
4	:set nu Displays lines with line numbers on the left side
5	:set sw Sets the width of a software tabstop. For example, you would set a shift width of 4 with this command — :set sw = 4
6	:set ws If <i>wraps</i> is set, and the word is not found at the bottom of the file, it will try searching for it at the beginning
7	:set wm If this option has a value greater than zero, the editor will automatically "word wrap". For example, to set the wrap margin to two characters, you would type this: :set wm = 2
8	:set ro Changes file type to "read only"
9	:set term Prints terminal type

10

:set bf

Discards control characters from input

Running Commands

The vi has the capability to run commands from within the editor. To run a command, you only need to go to the command mode and type **:!** command.

For example, if you want to check whether a file exists before you try to save your file with that filename, you can type **:! ls** and you will see the output of **ls** on the screen.

You can press any key (or the command's escape sequence) to return to your vi session.

Replacing Text

The substitution command (**:s/**) enables you to quickly replace words or groups of words within your files. Following is the syntax to replace text –

```
:s/search/replace/g
```

The **g** stands for globally. The result of this command is that all occurrences on the cursor's line are changed.

Important Points to Note

The following points will add to your success with vi –

- You must be in command mode to use the commands. (Press Esc twice at any time to ensure that you are in command mode.)
- You must be careful with the commands. These are case-sensitive.
- You must be in insert mode to enter text.

Basic UNIX commands

Note: not all of these are actually part of UNIX itself, and you may not find them on all UNIX machines. But they can all be used on **turing** in essentially the same way, by typing the command and hitting return. Note that some of these commands are different on non-Solaris machines - see [SunOS differences](#).

If you've made a typo, the easiest thing to do is hit **CTRL-u** to cancel the whole line. But you can also edit the command line (see the guide to [More UNIX](#)).

UNIX is case-sensitive.

Files

- **ls** --- lists your files
 - ls -l** --- lists your files in 'long format', which contains lots of useful information, e.g. the exact size of the file, who owns the file and who has the right to look at it, and when it was last modified.
 - ls -a** --- lists all files, including the ones whose filenames begin in a dot, which you do not always want to see.

There are many more options, for example to list files by size, by date, recursively etc.
- **more filename** --- shows the first part of a file, just as much as will fit on one screen. Just hit the space bar to see more or **q** to quit. You can use **/pattern** to search for a pattern.
- **emacs filename** --- is an editor that lets you create and edit a file. See the [emacs page](#).
- **mv filename1 filename2** --- moves a file (i.e. gives it a different name, or moves it into a different directory (see below))
- **cp filename1 filename2** --- copies a file
- **rm filename** --- removes a file. It is wise to use the option **rm -i**, which will ask you for confirmation before actually deleting anything. You can make this your default by making an [alias](#) in your **.cshrc** file.
- **diff filename1 filename2** --- compares files, and shows where they differ
- **wc filename** --- tells you how many lines, words, and characters there are in a file
- **chmod options filename** --- lets you change the read, write, and execute permissions on your files. The default is that only you can look at them and change them, but you may sometimes want to change these permissions. For example, **chmod o+r filename** will make the file readable for everyone, and **chmod o-r filename** will make it unreadable for others again. Note that for someone to be able to actually look at the file the directories it is in need to be at least executable. See [help protection](#) for more details.
- File Compression
 - **gzip filename** --- compresses files, so that they take up much less space. Usually text files compress to about half their original size, but it depends very much on the size of the file and the nature of the contents. There are other tools for this purpose, too (e.g. **compress**), but gzip usually gives the highest compression rate. Gzip produces files with the ending '.gz' appended to the original filename.
 - **gunzip filename** --- uncompresses files compressed by gzip.
 - **gzcat filename** --- lets you look at a gzipped file without actually having to gunzip it (same as **gunzip -c**). You can even print it directly, using **gzcat filename | lpr**
- printing
 - **lpr filename** --- print. Use the **-P** option to specify the printer name if you want to use a printer other than your default printer. For example, if you want to print double-sided, use

'lpr -Pvalkyr-d', or if you're at CSLI, you may want to use 'lpr -Pcord115-d'. See 'help printers' for more information about printers and their locations.

- **lpq** --- check out the printer queue, e.g. to get the number needed for removal, or to see how many other files will be printed before yours will come out
- **lprm *jobnumber*** --- remove something from the printer queue. You can find the job number by using lpq. Theoretically you also have to specify a printer name, but this isn't necessary as long as you use your default printer in the department.
- **genscript** --- converts plain text files into postscript for printing, and gives you some options for formatting. Consider making an alias like **alias ecop 'genscript -2 -r \!* | lpr -h -Pvalkyr'** to print two pages on one piece of paper.
- **dvips *filename*** --- print .dvi files (i.e. files produced by LaTeX). You can use **dviselect** to print only selected pages. See the [LaTeX page](#) for more information about how to save paper when printing drafts.

Directories

Directories, like folders on a Macintosh, are used to group files together in a hierarchical structure.

- **mkdir *dirname*** --- make a new directory
- **cd *dirname*** --- change directory. You basically 'go' to another directory, and you will see the files in that directory when you do 'ls'. You always start out in your 'home directory', and you can get back there by typing 'cd' without arguments. 'cd ..' will get you one level up from your current position. You don't have to walk along step by step - you can make big leaps or avoid walking around by specifying [pathnames](#).
- **pwd** --- tells you where you currently are.

Finding things

- **ff** --- find files anywhere on the system. This can be extremely useful if you've forgotten in which directory you put a file, but do remember the name. In fact, if you use **ff -p** you don't even need the full name, just the beginning. This can also be useful for finding other things on the system, e.g. documentation.
- **grep *string filename(s)*** --- looks for the string in the files. This can be useful a lot of purposes, e.g. finding the right file among many, figuring out which is the right version of something, and even doing serious corpus work. grep comes in several varieties (**grep**, **egrep**, and **fgrep**) and has a lot of very flexible options. Check out the man pages if this sounds good to you.

About other people

- **w** --- tells you who's logged in, and what they're doing. Especially useful: the 'idle' part. This allows you to see whether they're actually sitting there typing away at their keyboards right at the moment.
- **who** --- tells you who's logged on, and where they're coming from. Useful if you're looking for someone who's actually physically in the same building as you, or in some other particular location.

- **finger *username*** --- gives you lots of information about that user, e.g. when they last read their mail and whether they're logged in. Often people put other practical information, such as phone numbers and addresses, in a file called **.plan**. This information is also displayed by 'finger'.
- **last -1 *username*** --- tells you when the user last logged on and off and from where. Without any options, **last** will give you a list of everyone's logins.
- **talk *username*** --- lets you have a (typed) conversation with another user
- **write *username*** --- lets you exchange one-line messages with another user
- **elm** --- lets you send e-mail messages to people around the world (and, of course, read them). It's not the only mailer you can use, but the one we recommend. See the [elm page](#), and find out about the departmental [mailing lists](#) (which you can also find in /user/linguistics/helpfile).

About your (electronic) self

- **whoami** --- returns your username. Sounds useless, but isn't. You may need to find out who it is who forgot to log out somewhere, and make sure **you** have logged out.
- **finger & .plan files**
of course you can finger yourself, too. That can be useful e.g. as a quick check whether you got new mail. Try to create a useful .plan file soon. Look at other people's .plan files for ideas. The file needs to be readable for everyone in order to be visible through 'finger'. Do 'chmod a+r .plan' if necessary. You should realize that this information is accessible from anywhere in the world, not just to other people on turing.
- **passwd** --- lets you change your password, which you should do regularly (at least once a year). See the [LRB guide](#) and/or look at [help password](#).
- **ps -u *yourusername*** --- lists your processes. Contains lots of information about them, including the process ID, which you need if you have to kill a process. Normally, when you have been kicked out of a dialin session or have otherwise managed to get yourself disconnected abruptly, this list will contain the processes you need to kill. Those may include the shell (tcsh or whatever you're using), and anything you were running, for example emacs or elm. Be careful not to kill your current shell - the one with the number closer to the one of the ps command you're currently running. But if it happens, don't panic. Just try again :) If you're using an X-display you may have to kill some X processes before you can start them again. These will show only when you use **ps -efl**, because they're root processes.
- **kill *PID*** --- kills (ends) the processes with the ID you gave. This works only for your own processes, of course. Get the ID by using **ps**. If the process doesn't 'die' properly, use the option -9. But attempt without that option first, because it doesn't give the process a chance to finish possibly important business before dying. You may need to kill processes for example if your modem connection was interrupted and you didn't get logged out properly, which sometimes happens.
- **quota -v** --- show what your disk quota is (i.e. how much space you have to store files), how much you're actually using, and in case you've exceeded your quota (which you'll be given an automatic warning about by the system) how much time you have left to sort them out (by deleting or gzipping some, or moving them to your own computer).
- **du *filename*** --- shows the disk usage of the files and directories in *filename* (without argument the current directory is used). **du -s** gives only a total.
- **last *yourusername*** --- lists your last logins. Can be a useful memory aid for when you were where, how long you've been working for, and keeping track of your phonebill if you're making a non-local phonecall for dialling in.

Connecting to the outside world

- **nn** --- allows you to read news. It will first let you read the news local to turing, and then the remote news. If you want to read only the local or remote news, you can use **nnl** or **nnr**, respectively. To learn more about **nn** type **nn**, then `\tty{:man}`, then `\tty{=. *}`, then `\tty{Z}`, then hit the space bar to step through the manual. Or look at the man page. Or check out the [hypertext nn FAQ](#) - probably the easiest and most fun way to go.
- **rlogin hostname** --- lets you connect to a remote host
- **telnet hostname** --- also lets you connect to a remote host. Use **rlogin** whenever possible.
- **ftp hostname** --- lets you download files from a remote host which is set up as an ftp-server. This is a common method for exchanging academic papers and drafts. If you need to make a paper of yours available in this way, you can (temporarily) put a copy in /user/ftp/pub/TMP. For more permanent solutions, ask Emma. The most important commands within ftp are **get** for getting files from the remote machine, and **put** for putting them there (**mget** and **mput** let you specify more than one file at once). Sounds straightforward, but be sure not to confuse the two, especially when your physical location doesn't correspond to the direction of the ftp connection you're making. ftp just overwrites files with the same filename. If you're transferring anything other than ASCII text, use binary mode.
- **lynx** --- lets you browse the web from an ordinary terminal. Of course you can see only the text, not the pictures. You can type any URL as an argument to the **G** command. When you're doing this from any Stanford host you can leave out the **.stanford.edu** part of the URL when connecting to Stanford URLs. Type **H** at any time to learn more about **lynx**, and **Q** to exit.

Miscellaneous tools

- **webster word** --- looks up the word in an electronic version of Webster's dictionary and returns the definition(s)
- **date** --- shows the current date and time.
- **cal** --- shows a calendar of the current month. Use e.g., 'cal 10 1995' to get that for October 95, or 'cal 1995' to get the whole year.

Unix - Useful Commands

This quick guide lists commands, including a syntax and a brief description. For more detail, use –

\$man command

Files and Directories

These commands allow you to create directories and handle files.

Given below is the list of commands in Files and Directories.

Manipulating data

The contents of files can be compared and altered with the following commands.

Given below is the list of commands in Manipulating data.

Compressed Files

Files may be compressed to save space. Compressed files can be created and examined.

S.No.	Command & Description
1	compress Compresses files
2	gunzip Helps uncompress gzipped files
3	gzip GNU alternative compression method
4	uncompress Helps uncompress files
5	unzip

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	List, test and extract compressed files in a ZIP archive
6	zcat Cat a compressed file
7	zcmp Compares compressed files
8	zdiff Compares compressed files
9	zmore File perusal filter for crt viewing of compressed text

Getting Information

Various Unix manuals and documentation are available on-line. The following Shell commands give information –

S.No.	Command & Description
1	apropos Locates commands by keyword lookup
2	info Displays command information pages online
2	man Displays manual pages online

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3	whatis Searches the whatis database for complete words
4	yelp GNOME help viewer

Network Communication

These following commands are used to send and receive files from a local Unix hosts to the remote host around the world.

S.No.	Command & Description
1	ftp File transfer program
2	rcp Remote file copy
3	rlogin Remote login to a Unix host
4	rsh Remote shell
5	tftp Trivial file transfer program
6	telnet

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	Makes terminal connection to another host
7	ssh Secures shell terminal or command connection
8	scp Secures shell remote file copy
9	sftp Secures shell file transfer program

Some of these commands may be restricted at your computer for security reasons.

Messages between Users

The Unix systems support on-screen messages to other users and world-wide electronic mail –

S.No.	Command & Description
1	evolution GUI mail handling tool on Linux
2	mail Simple send or read mail program
3	mesg Permits or denies messages
4	parcel Sends files to another user

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5	pine Vdu-based mail utility
6	talk Talks to another user
7	write Writes message to another user

Programming Utilities

The following programming tools and languages are available based on what you have installed on your Unix.

Given below is the list of tools and languages in Programming Utilities.

Misc Commands

These commands list or alter information about the system –

Given below is the list of Misc Commands in Unix.

S.No.	Command & Description
1	chfn Changes your finger information
2	chgrp Changes the group ownership of a file
3	chown Changes owner

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4	date Prints the date
5	determin Automatically finds terminal type
6	du Prints amount of disk usage
7	echo Echo arguments to the standard options
8	exit Quits the system
9	finger Prints information about logged-in users
10	groupadd Creates a user group
11	groups Show group memberships
12	homequota Shows quota and file usage

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13	iostat Reports I/O statistics
14	kill Sends a signal to a process
15	last Shows last logins of users
16	logout Logs off Unix
17	lun Lists user names or login ID
18	netstat Shows network status
19	passwd Changes user password
20	passwd Changes your login password
21	printenv Displays value of a shell variable

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22	ps Displays the status of current processes
23	ps Prints process status statistics
24	quota -v Displays disk usage and limits
25	reset Resets terminal mode
26	script Keeps script of terminal session
27	script Saves the output of a command or process
28	setenv Sets environment variables
30	stty Sets terminal options
31	time Helps time a command

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32	top Displays all system processes
33	tset Sets terminal mode
34	tty Prints current terminal name
35	umask Show the permissions that are given to view files by default
36	uname Displays name of the current system
37	uptime Gets the system up time
38	useradd Creates a user account
39	users Prints names of logged in users
40	vmstat Reports virtual memory statistics

41	w Shows what logged in users are doing
42	who Lists logged in users